



Anne Yilimu Avnsbjerg Lundkvist

HAR DEN 29. JANUAR 2025 BESTÅET

Diplomingeniøruddannelsen i sundhedsteknologi

VED AARHUS UNIVERSITET
OG HAR DERMED RET TIL AT BETEGNE SIG

Professionsbachelor som Diplomingeniør i
sundhedsteknologi
Bachelor of Engineering in Healthcare Technology

Aarhus, den 31. januar 2025

Eskild Holm Nielsen

dekan for Aarhus Universitet Faculty of Technical Sciences





I henhold til bekendtgørelse nr. 2674 af 28. december 2021 kvalificerer uddannelsen til professionsbachelor som diplomingeniør de studerende til nationalt og internationalt at varetage erhvervsfunktioner, hvor de skal omsætte tekniske forskningsresultater samt naturvidenskabelig og teknisk viden til praktisk anvendelse ved udviklingsopgaver og ved løsning af tekniske problemer. Herudover kan de kritisk tilegne sig ny viden inden for relevante ingeniørmæssige områder og selvstændigt løse forekommende ingeniørmæssige arbejdsopgaver.

Endelig kan de planlægge, realisere og styre tekniske og teknologiske anlæg og herunder være i stand til at inddrage samfundsmæssige, økonomiske, miljø- og arbejdsmiljømæssige konsekvenser i løsningen af tekniske problemer samt indgå i samarbejds- og ledelsesmæssige funktioner og sammenhænge på et kvalificeret niveau sammen med mennesker, der har forskellig uddannelsesmæssig, sproglig og kulturel baggrund.

Uddannelsen kvalificerer herudover de studerende til at deltage i videreuddannelse.

Diplomingeniøruddannelsen i sundhedsteknologi er normeret til: **210 ECTS**

Anne Yilimu Avnsbjerg Lundkvist

har opnået følgende resultater:

	<u>7-trinsskala</u>	<u>ECTS-skala</u>	<u>Bestået</u>
Obligatorisk del			
ST1IKLT-01 Indledende kredsløbsteknik 5 ECTS	02	E	Bestået
ST1ITS1-03 Programmering 1 5 ECTS	02	E	Bestået
ST1PRJ1-04 Semesterprojekt 1: Puls måler 5 ECTS			Bestået
ST1MMLS-01 Matematisk modellering af lineære systemer 5 ECTS	12	A	Bestået
ST1SUN1-02 Sundhedsvidenskab 1 10 ECTS	7	C	Bestået
ST1VPR1-01 Værkstedspraktik 1			Bestået
ST2ASA-01 Anvendt analog systemanalyse 10 ECTS	10	B	Bestået
ST2ISE-01 Indledende system engineering 5 ECTS	10	B	Bestået
ST2ITS2-02 Programmering 2 5 ECTS			Bestået
ST2PRJ2-05 Semesterprojekt 2: EKG diagnostik 5 ECTS			Bestået
ST2SUN2-03 Sundhedsvidenskab 2 5 ECTS	10	B	Bestået
ST3DSB-01 Digital signalbehandling 5 ECTS			Bestået
ST3ITS3-04 Programmering 3 5 ECTS	4	D	Bestået
ST3KSS-02 Kommunikation og samarbejde i sundhedsvæsenet 5 ECTS	10	B	Bestået





	<u>7-trinsskala</u>	<u>ECTS-skala</u>	<u>Bestået</u>
ST3KVI-01 Kardiovaskulær instrumentering 5 ECTS	7	C	Bestået
ST3NVK-01 Netværkskommunikation 5 ECTS			Bestået
ST3PRJ3-03 Semesterprojekt 3: Udvikling af et blodtrykmålesystem 5 ECTS	10	B	Bestået
ST4DMD-02 Development of Medical Devices 5 ECTS	10	B	Bestået
ST4EFI-01 Elektrofysiologi og instrumentering 5 ECTS	12	A	Bestået
ST4MTV-02 Medicinsk teknologi vurdering 10 ECTS	12	A	Bestået
ST4PRJ4-01 Semesterprojekt 4 5 ECTS			Bestået
ST4SWT-02 Software test 5 ECTS	12	A	Bestået
Ingeniørpraktik E5IPR-01 Ingeniørpraktik 30 ECTS			Bestået
Specialisering ETPRJ-01 Project Work - Study Course 5 ECTS			Bestået
STTBML-01 Biomedical Machine Learning 5 ECTS	12	A	Bestået
STTPCO-01 Pervasive Computing 5 ECTS	10	B	Bestået
STTUX-01 UX Design 5 ECTS			Bestået
SW4BAD-01 Back-end udvikling og databaser 10 ECTS			Bestået
SW4FED-02 Front-end udvikling 5 ECTS	7	C	Bestået
SWMAD-01 Mobile Application Development 5 ECTS			Bestået





ST7BAC-01 Bachelorprojekt

20 ECTS
App til nyfødtes trivsel

7-trinsskala ECTS-skala Bestået

12 A Bestået

Bevisets rigtighed bekræftes

Aarhus, den 31. januar 2025

Louise Westh Møller
uddannelsesadministrativ medarbejder





Kompetenceprofil for uddannelsen

Viden

- har grundlæggende viden inden for matematik, fysiologi og signalbehandling.
- har grundlæggende viden inden for analog og digital elektronik.
- har grundlæggende viden om sundhedsvæsenets opbygning og problemstillinger.
- har udviklingsbaseret viden inden for instrumentering, systematiske udviklingsprocesser, evidensbaseret udvikling, objektorienteret programmering og softwareteknologier.
- kan forstå og reflektere over faglig viden i relation til medicinsk systemudvikling.
- har udviklingsbaseret viden om brugerinddragelse og innovation relevant for medicinsk systemudvikling.

Færdigheder

- kan anvende sin faglige viden om kommunikation og brugerinddragelse i udviklingen af nye, innovative løsninger til sundhedssystemet.
- kan på baggrund af sin faglige viden, udvælge den optimale teknologiske løsning til en konkret problemstilling og argumentere for valget af denne.
- kan udarbejde et relevant løsningsforslag til en konkret problemstilling, implementere et løsningsforslag og sætte dette i drift.
- kan formidle praksisnære og faglige problemstillinger og løsningsmuligheder til samarbejdspartnere og brugere både inden for og uden for sundhedsvæsenet.

Kompetencer

- kan arbejde udviklings- og innovationsorienteret i forbindelse med sundhedsrelevante problemstillinger.
- kan foretage proces- og projektstyring i forbindelse med udviklingsprojekter.
- kan samarbejde professionelt og etisk ansvarligt med fagfæller og med andre interessenter.
- kan i forbindelse med nye opgaver og problemstillinger tilegne sig ny viden, nye færdigheder og nye kompetencer i relation til medicinsk system- og softwareudvikling.





Anne Yilimu Avnsbjerg Lundkvist

HAS ON 29 JANUARY 2025 PASSED THE EXAMINATIONS REQUIRED FOR THE

Bachelor's Degree Programme in Healthcare Technology

AT AARHUS UNIVERSITY
AND HAS THUS BEEN AWARDED THE DEGREE

Bachelor of Engineering in Healthcare Technology
Professionsbachelor som Diplomingeniør i sundhedsteknologi

Aarhus, 31 January 2025

Eskild Holm Nielsen

Dean of Aarhus University Faculty of Technical Sciences





Pursuant to Ministerial Order no. 2674 of 28 December 2021, the Bachelor's Degree Programme qualifies graduates to positions in industry, nationally as well as internationally, where they will apply technology research results and scientific and technological knowledge to development work and to solving technical problems.

In addition, they will be able to take a critical approach to the acquisition of new knowledge in relevant aspects of the field of engineering and independently carry out engineering work.

Finally, they will be able to plan, implement and manage technical and technological plants and projects, which includes being able to integrate social, economic, environmental and work environment-related consequences in the solution of technical problems. They will also be able to participate in collaborative and managerial positions and contexts on a highly qualified level with people of different educational, linguistic and cultural backgrounds.

The degree programme will furthermore qualify graduates for postgraduate studies.

The Bachelor's Degree Programme in Healthcare Technology is rated at: **210 ECTS**

Anne Yilimu Avnsbjerg Lundkvist

has obtained the following results:

	<u>7-point scale</u>	<u>ECTS scale</u>	<u>Passed</u>
Compulsory Part			
ST1IKLT-01 Introduction to Electric Circuit Theory 5 ECTS	02	E	Passed
ST1ITS1-03 Programming 1 5 ECTS	02	E	Passed
ST1PRJ1-04 Project 1: Heart Rate Monitor 5 ECTS			Passed
ST1MMLS-01 Mathematical Modeling of Linear Systems 5 ECTS	12	A	Passed
ST1SUN1-02 Health Science 1 10 ECTS	7	C	Passed
ST1VPR1-01 Practical Training 1			Passed
ST2ASA-01 Applied Analogue System Analysis 10 ECTS	10	B	Passed
ST2ISE-01 Introduction to System Engineering 5 ECTS	10	B	Passed
ST2ITS2-02 Programming 2 5 ECTS			Passed
ST2PRJ2-05 Project 2: ECG Diagnostics 5 ECTS			Passed
ST2SUN2-03 Health Science 2 5 ECTS	10	B	Passed
ST3DSB-01 Digital Signalprocessing 5 ECTS			Passed
ST3ITS3-04 Programming 3 5 ECTS	4	D	Passed
ST3KSS-02 Communication and Cooperation in the Healthcare System 5 ECTS	10	B	Passed



301098-2056
201909232

	<u>7-point scale</u>	<u>ECTS scale</u>	<u>Passed</u>
ST3KVI-01 Cardiovascular Instrumentation 5 ECTS	7	C	Passed
ST3NVK-01 Network Communication 5 ECTS			Passed
ST3PRJ3-03 Project 3: Development of Bloodpressure Monitor System 5 ECTS	10	B	Passed
ST4DMD-02 Development of Medical Devices 5 ECTS	10	B	Passed
ST4EFI-01 Electrophysiology and Instrumentation 5 ECTS	12	A	Passed
ST4MTV-02 Health Technology Assessment 10 ECTS	12	A	Passed
ST4PRJ4-01 Semester Project 4 5 ECTS			Passed
ST4SWT-02 Software Test 5 ECTS	12	A	Passed
Internship E5IPR-01 Internship 30 ECTS			Passed
Specialisation ETPRJ-01 Project Work - Study Course 5 ECTS			Passed
STTBML-01 Biomedical Machine Learning 5 ECTS	12	A	Passed
STTPCO-01 Pervasive Computing 5 ECTS	10	B	Passed
STTUX-01 UX Design 5 ECTS			Passed
SW4BAD-01 Back-end Development and Databases 10 ECTS			Passed
SW4FED-02 Front-end Development 5 ECTS	7	C	Passed
SWMAD-01 Mobile Application Development 5 ECTS			Passed





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ST7BAC-01 Bachelor's Project

20 ECTS
App for newborn well-being

7-point scale ECTS scale Passed

12 A Passed

The validity of this document is confirmed

Aarhus, 31 January 2025

Louise Westh Møller
Administrative Officer





Skills Profile for the Programme

Knowledge

- has basic knowledge of mathematics, physiology and signal processing.
- has basic knowledge of analogue and digital electronics.
- has basic knowledge of the structure and issues in the healthcare system.
- has development-based knowledge of instrumentation, systematic development processes, evidence-based development, object-oriented programming and software technologies.
- can understand and reflect on academic knowledge in relation to medical system development.
- has development-based knowledge of user involvement and innovation relevant to medical system development.

Skills

- is able to apply knowledge about communication and user involvement to develop new innovative solutions for the healthcare system.
- is able to apply knowledge to choose the optimal technological solution to a specific problem and argue in favour of this choice.
- is able to propose a solution to a specific problem, implement the proposed solution and put it into operation.
- is able to communicate practice-based and specialist issues and solutions to collaboration partners and users, both within and outside the healthcare system.

Competences

- is able to take an innovative and development-oriented approach in connection with health-related issues.
- is able to carry out process and project management in connection with development projects.
- is able to collaborate professionally and ethically with peers and with other stakeholders.
- is able to acquire new knowledge, new skills and new competences in relation to medical system and software development in connection with new tasks and issues.





Diploma Supplement

This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1 INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

1.1 Last name(s)

Lundkvist

1.2 First name(s)

Anne Yilimu Avnsbjerg

1.3 Date of birth (dd/mm/yyyy)

30/10/1998

1.4 Student identification number or code

Matriculation number: 201909232 / Civil registration number: 301098-2056

2 INFORMATION IDENTIFYING THE QUALIFICATION

2.1 Name of qualification and (if applicable) title conferred (in original language)

Professionsbachelor som Diplomingeniør i sundhedsteknologi
Bachelor of Engineering in Healthcare Technology

2.2 Main field(s) of study for the qualification

Analog Electronics, Medical Instrumentation, Mathematics, Signal Processing, Programming Languages and Principles, Communication Systems and Networks, Information Technology, Human Biology, Communication and Collaboration with the Healthcare System.

2.3 Name and status of awarding institution (in original language)

Aarhus Universitet (Aarhus University) is an independent institution under the public-sector administration and supervised by the Ministry of Higher Education and Science and regulated according to the University Act with subsequent changes.

Aarhus Universitet (Aarhus University) is a university that has undergone external quality assurance by the Danish Accreditation Institution (in Danish: Danmarks Akkrediteringsinstitution), that is certified to follow the European Standards and Guidelines through registration in EQAR and membership in ENQA, in Denmark.

2.4 Name and status of institution (if different from 2.3) administering studies (in original language)

Not applicable / as above





2.5 Language(s) of instruction/examination

Teaching/examination at Aarhus University takes place in Danish and English, although other languages may be used when appropriate.

3 INFORMATION ON THE LEVEL AND DURATION OF THE QUALIFICATION

3.1 Level of qualification

Bachelor's degree at NQF/EQF Level 6 referring to First Cycle in the Bologna QF

3.2 Official duration of programme in credits and/or years

210 ECTS

3.3 Access requirements

Admission to the Bachelor's Degree Programme requires a) The Upper Secondary School Leaving Examination, or b) Higher Preparatory Examination, or c) Higher Business Examination or d) Higher Technical Examination. Applicants with other qualifications may be admitted after an assessment of their qualifications.

4 INFORMATION ON THE PROGRAMME COMPLETED AND THE RESULTS OBTAINED

4.1 Mode of study

Full-time.

4.2 Programme learning outcomes

The programme aims at educating healthcare technology engineers of high professional standard. During the first four semesters, students complete mandatory courses in Mathematics, Programming, Health Science, Electric Circuit Theory, Signal Processing, Biomechanics, Health Technology Assessment, Communication and Collaboration with the Healthcare System. The fifth semester is comprised of an internship in public or private health care or with an industry partner. During the sixth and seventh semester, students complete additional mandatory and elective courses, allowing them to develop a subject area concentration. In the seventh semester, a Bachelor project is carried out, generally in collaboration with an industry partner. Graduates are able to apply scientific research and technological knowledge to engineering problems, allowing them to independently plan, perform and evaluate engineering tasks with a combination of practical, technical and management skills. They are qualified to work nationally as well as internationally in the field of engineering, in both public and private sector health care industry.

4.3 Programme details, individual credits gained and grades/marks obtained

Please refer to the enclosed transcript of records

4.4 Grading system and, if available, grade distribution table

<http://ufm.dk/en/education-and-institutions/the-danish-education-system/grading-system>

4.5 Overall classification of the qualification (in the original language)

Not applicable for Danish qualifications





5 INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1 Access to further study

A completed Bachelor's degree at NQF/EQF Level 6 referring to First Cycle in the Bologna QF gives access further study within relevant fields at NQF/EQF Level 7 referring to Second Cycle in the Bologna QF. Specific admission requirements may be established by receiving programmes.

5.2 Access to a regulated profession (if applicable)

Graduates of the programme are entitled to use the title Bachelor of Engineering in Healthcare Technology. They are eligible for membership in IDA, Ingeniørforeningen i Danmark, the professional organization for engineers in Denmark.

6 ADDITIONAL INFORMATION

6.1 Additional information

Aarhus University offers unique, alternative opportunities for research and education cutting across many different subjects, for the benefit of both students and researchers, as well as the authorities and the business community. These interdisciplinary combinations provide exceptional opportunities. Aarhus University combines quality in its services with diversity - a diversity that also makes sure that the university is in wide-reaching contact with all the important sectors of society.

Aarhus University has an international focus and makes targeted efforts to attract researchers and students from abroad.

Research and education

Academic values form the basis for all activities at Aarhus University. Via curious research, critical analysis and ongoing debate, researchers and students endeavour to find new ways to gain insight, understanding and education for the benefit of society as a whole. The university consists of four main academic areas. Combined, they cover the entire research spectrum - basic research, applied research, strategic research and research-based advice to the authorities. In all degree programmes, research and education are closely connected, and the research-based instruction - including teaching that spans the main academic areas - ensures the depth of the degree programmes.

A visionary university

The mission of Aarhus University is to ensure and develop knowledge, welfare and culture through research and research-based education, knowledge dissemination and external advice. The vision of Aarhus University is to belong to the elite of universities and to contribute to the development of national and global welfare via outstanding research and world-class degree programmes. The values of Aarhus University are based on the ethical challenges regarding freedom and independence that are described in the Magna Charta of European Universities. Staff and students at Aarhus University work enquiringly and critically, in open and dynamic interaction with the surrounding world.

6.2 Further information sources

For further information on this degree programme, please refer to <http://studieguide.au.dk/en> and the Aarhus University web site <http://www.au.dk/en>.





7 CERTIFICATION OF THE SUPPLEMENT

7.1 Date

31 January 2025

7.2 Signature

Eskild Holm Nielsen

7.3 Capacity

Dean of Aarhus University Faculty of Technical Sciences

7.4 Official stamp or seal



8 INFORMATION ON THE NATIONAL HIGHER EDUCATION SYSTEM

Please see the attached description of The Danish Higher Education System of April 2016

